

Weitong Li

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EDUCATION

Ph.D. Computer Science and Engineering, Virginia Tech, Dec. 2025
B.Eng. Software Engineering, Zhejiang University, Sept. 2020

WORKING EXPERIENCE

Dec. 2025 - Current Postdoctoral Researcher, Virginia Tech
May - Aug. 2024 Research Intern, Cablelabs
May - Aug. 2023 Research Intern, Cloudflare

RESEARCH AREAS

My research focuses on measuring, understanding, and securing Internet infrastructure. I conduct Internet-scale measurements to uncover real-world behaviors, risks, and dependencies, and design AI-driven systems to improve the security, performance, and resilience of networked systems.

PUBLICATIONS

(*: equal contribution)

Refereed Conference Publications

- 2026 W. Li, Y. Xu, T. Chung “The Threat Landscape of IP Leasing in the Post-RPKI Era.” In Proceedings of the IEEE Symposium on Security and Privacy (**Oakland’26**), San Francisco, USA, May 2026
- 2026 W. Li*, Y. Xu*, M. Zhang, V. Giotsas, T. Chung “RScope: Unveiling Global ROV Deployments and Dependencies in the Post-ROV Era” In Proceedings of the ACM Internet Measurement Conference (**IMC’26**), Karlsruhe, Germany, October 2026
- 2026 W. Li, T. Wan, T. Chung “Demystifying RPKI-Invalid Prefixes: Hidden Causes and Security Risks.” In Proceedings of the Network and Distributed System Security Symposium (**NDSS’26**), San Diego, USA, February 2026
- 2025 W. Li, Y. Li, T. Chung “ImpROV: Measurement and Practical Mitigation of Collateral Damage of RPKI Route Origin Validation” In Proceedings of the USENIX Security Symposium (**Security’25**), Seattle, USA, August 2025
- 2024 M. Khan, W. Li, T. Fiebig, T. Chung “SPF Beyond the Standard: Management and Operational Challenges in Practice and Practical Recommendations.” In Proceedings of the USENIX Security Symposium (**Security’24**), Philadelphia, USA, August 2024
- 2024 M. Kang, W. Li, R. Rijswijk-Deij, T. Kwon, T. Chung “IRRedicator: Pruning IRR with RPKI-Valid BGP Insights.” In Proceedings of the Network and Distributed System Security Symposium (**NDSS’24**), San Diego, USA, February 2024

- 2023 W. Li, Z. Lin, M. Khan, R. Fontugne, E. Aben, T. Chung “RoVista: Measuring and Understanding the Route Origin Validation (ROV) in RPKI.” In Proceedings of the ACM Internet Measurement Conference (**IMC’23**), Montreal, Canada, October 2023
- 2023 M. Khan, W. Li, T. Fiebig, T. Chung “Youve Got Report: Measurement and Security Implications of DMARC Reporting.” In Proceedings of the USENIX Security Symposium (**Security’23**), Anaheim, USA, August 2023

Under Submission

- 2026 Y. Kim, S. Baek, W. Li, T. Chung, M. S. Kang “iROV: Breaking the Silence of RPKI with Interactive Validation.” Under Submission.
- 2026 W. Li, Y. Xu, T. Chung “Abuse Reporting in a Decoupled Internet: The Structural Gap Between Registry and Operation.” Under Submission.
- 2026 W. Li, Y. Li, T. Chung “pqRPKI: A Practical Hash-based RPKI Architecture for the Post-Quantum Era.” Under Submission (arXiv:2603.06968).
- 2026 Y. Xu*, W. Li*, E. Umrani, T. Chung “ASINT: Learning AS-to-Organization Mapping from Internet Metadata” Under Submission (arXiv:2508.02571).

INVITED TALKS

- 2026 “When RPKI Meets Reality: Understanding and Improving Internet Routing Security” Job talks at Tsinghua, Xiamen, HUST, SEU, WFU, and some other Universities.
- 2025 “The Threat Landscape of IP Leasing in the RPKI-Era.” IRTF-maprg (IETF 124).
- 2025 “Progress Yet to Be Fulfilled: Measuring Collateral Damage in ROV.” IRTF-maprg (IETF 122).
- 2024 “Measuring and Securing RPKI.” Tsinghua University.
- 2023 “Measurement and Security Implications of DMARC Reporting.” Tsinghua University.
- 2023 “RoVista: Measuring and Analyzing the Route Origin Validation.” IRTF-maprg (IETF 118).

GRANTS AND AWARDS

- 2026 ARIN Fellow for ARIN57
- 2026 ISOC Pulse Fellowship
- 2025 Usenix Travel Grant (\$1,500)
- 2025 IRTF Travel Grant (\$2,500)
- 2024 Pratt Fellowship, Computer Science Department, Virginia Tech
- 2017 Cyrus Tang Scholarship, Zhejiang University

SERVICE

TPC Member

- 2026 Network Traffic Measurement and Analysis Conference (TMA)
- 2026 ACM Conference on Computer and Communications Security (CCS)
- 2026 USENIX Security Symposium (Security)
- 2022,25 Shadow TPC for ACM Internet Measurement Conference (IMC)

Reviewer

IEEE Transactions on Dependable and Secure Computing (TDSC)

IEEE Transactions on Information Forensics and Security (TIFS)

IEEE/ACM Transactions on Networking (ToN)

IEEE Internet of Things Journal (IoT-J)

Computer Networks (CN)

REFERENCE

Taejoong (Tijay) Chung, Associate Professor, Virginia Tech, tijay@vt.edu

Danfeng (Daphne) Yao, Professor, Virginia Tech, danfeng@vt.edu

Tao Wan, Distinguished Technologist, Cablelabs, t.wan@cablelabs.com

Vasilis Giotsas, Research Engineer, Cloudflare Research, vasilis@cloudflare.com

Xiaohu Yang, Professor, Zhejiang University, yangxh@zju.edu.cn